Modelling protein synthesis

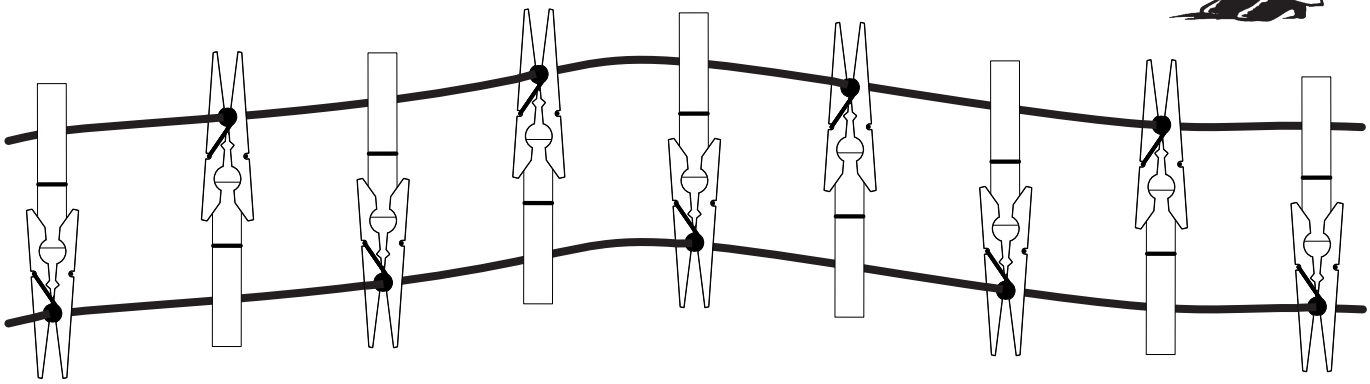
Recount



Yesterday Mrs De-enai asked us to investigate protein synthesis in cells. To do this, we constructed a model to show us how proteins are made. Each group was given a set of plastic pegs of five different colours – blue, green, yellow, red and white, wool in different colours, markers and paper.

Step 1

First we had to model transcription in the nucleus. To do this, we constructed a double strand of DNA. Our group decided that the red pegs would be adenine, blue pegs for thymine, yellow for guanine and green for cytosine. We threaded black coloured wool through the spaces in the pegs using the sequence CTTAAGATG, corresponding to the colours we had chosen. We then made a complementary strand of the DNA and pegged the two strands together. Refer to the diagram below.



1 Write the colours of the pegs in the original DNA strand. Make sure the order of the colours is correct. Then colour the pegs to show this order.

Step 2

The next step was to unzip the double-stranded DNA molecule, producing two single-strand molecules. Using our original strand, we made a complementary molecule but this time we used red wool instead of black wool, and instead of thymine, our blue peg, we used uracil, a white peg. So we joined uracil, not thymine, to adenine. We called this complementary strand messenger RNA (m-RNA).

2 Draw, colour and label a diagram to show the m-RNA molecule.

Step 3

In this part of our investigation, we modelled translation. Translation occurs in the cytoplasm of a cell. The m-RNA unzips from the DNA and leaves the nucleus and attaches itself on a ribosome. Using the third coloured wool, green, we made three transfer RNA (t-RNA) molecules: CUU AAG AUG

We attached a small piece of cardboard, labelled with the amino acids' names in the correct order: Leucine, Lysine, Methionine. The three t-RNA molecules were attached to the corresponding m-RNA molecule.

© Suada Bilali, Chris Greef and Blake Education 2010

Modelling protein synthesis

3 Draw, colour and label a diagram to show the three t-RNA molecules attached to the m-RNA molecule.

We stuck the amino acids together with tape and removed the t-RNA pegs. The end result was a polypeptide chain made up of all the amino acids.

4 Draw, colour and label a diagram to show the polypeptide chain made up of all the amino acids.

5 Write down the original DNA sequence._____

6 What was the complementary m-RNA sequence?_____

7 What were the t-RNA codes?_____

8 Describe in your own words what you have learned from this model._____

_____

9 Where does transcription take place?_____

10 Explain what happens to DNA during transcription._____

_____

11 Where does translation take place?_____

12 Explain what happens during translation._____

_____

13 The code on a section of DNA reads ATCGGCATATCTACT.

a What would the complementary m-RNA read? That is, what are the m-RNA codons?_____

b What would the anti-codons on t-RNA read?_____

14 Rewrite the recount as a procedural text type. Start with a heading, then include an aim, equipment list, method, results and a conclusion.

15 Use different coloured pegs and wool to model protein synthesis for yourself.

a Does this model help you understand protein synthesis? Explain.

b What are the limitations of this model?

c Suggest how you could improve this model of protein synthesis.

© Suada Bilali, Chris Greef and Blake Education 2010